

Mini-project CAM C301 Week 9 & 10: Forecasting TimeSeries Data

Rodrigo Grandy, PhD

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1 Introduction

In this report we present a comprehensive time series analysis of book sales data from Nielsen BookScan, focusing on sales forecasting for independent publishers. The analysis examines historical sales patterns for two prominent titles: “The Very Hungry Caterpillar” and “The Alchemist,” using data from 2012 onwards. Multiple forecasting methodologies are implemented and compared, including classical ARIMA models, machine learning approaches (light XGBoost), deep learning techniques (LSTM), and hybrid model combinations. The objective is to develop accurate sales forecasts to support publishers’ inventory management, stock control, and investment decisions. Performance evaluation utilizes statistical metrics including Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE) to assess forecasting accuracy across different methodological approaches.

2 Objectives

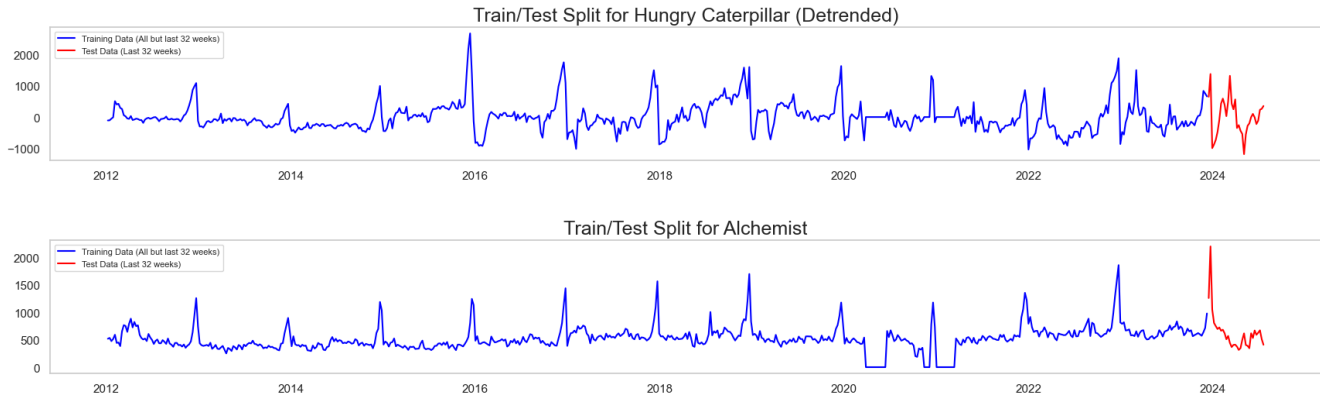
- Data Preprocessing: Clean and resample the sales data to create consistent weekly intervals for time series analysis.
- Time Series Pattern Analysis: Decompose sales data and test stationarity to understand seasonal trends and temporal characteristics.
- Apply Auto ARIMA models to forecast sales periods for both selected books.
- Implement XGBoost or lightGBM and LSTM models with hyperparameter tuning for sales forecasting.
- Hybrid Approaches: Combine SARIMA-LSTM models and compare weekly versus monthly prediction strategies.

3 Methods

The analysis utilized Python 3.11 with key libraries including pandas, numpy, statsmodels, scikit-learn, LightGBM, TensorFlow/Keras, and sktime for comprehensive time series forecasting on Nielsen BookScan data (2012-2024). Data preprocessing involved loading four Excel tabs, converting ISBNs to strings,

establishing datetime indices, and resampling to weekly intervals (W-SAT frequency) with zero-filling for missing periods. Post-2012 data filtering focused on “The Alchemist” and “The Very Hungry Caterpillar” for detailed analysis.

A



B

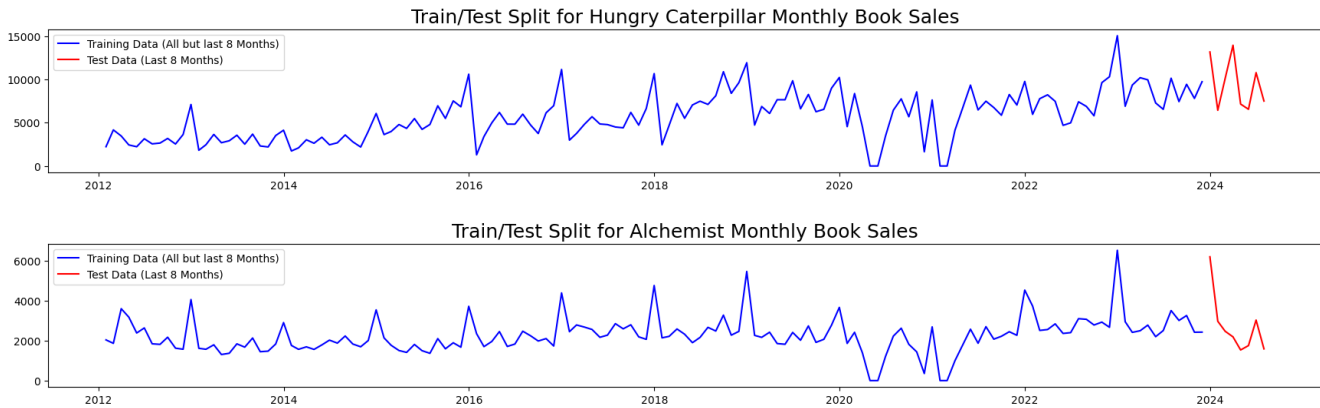


Figure 1: Train/Test Split for Weekly and Monthly Book Sales Data for both The Very Hungry Caterpillar and The Alchemist. A) Training and test data split for weekly sales series displayed from 2012 to 2024. B) Training and test data split for monthly sales series displayed from 2012 to 2024. Blue line represents the training data and the red line the test data corresponding to the last 32 weeks or 8 months.

Statistical preprocessing included stationarity testing (ADF, KPSS), ACF/PACF analysis, and STL decomposition. Classical forecasting employed Auto-ARIMA with seasonal parameters ($m=52$). Machine learning implementation featured LightGBM with recursive windowing, incorporating TransformedTargetForecaster for deseasonalization and detrending. Deep learning utilized multi-step LSTM architectures with MinMax scaling and various preprocessing methods. Hybrid models combined SARIMA-LSTM in sequential/parallel configurations.

Performance evaluation used 32-week out-of-sample forecasts, assessed via MAE and MAPE across weekly and monthly aggregations.

4 Results

Following initial EDA, both datasets were partitioned into training and test sets using temporal splits. Training data encompasses historical periods while test data represents the final 32 weeks (weekly series) and 8 months (monthly series). This chronological partitioning maintains time series integrity and enables robust out-of-sample forecasting evaluation, mimicking real-world prediction scenarios (Figure 1).

Initial statistical analysis assessed structural characteristics of both datasets. The Hungry Caterpillar exhibited strong seasonality (0.67) but weak trend (0.18), while The Alchemist showed both strong seasonality (0.61) and trend (0.69). Stationarity tests revealed The Alchemist was stationary and ready for modeling, whereas The Hungry Caterpillar required detrending due to trend-stationarity (See notebook2 for details). ACF/PACF analysis confirmed significant seasonal patterns at 52-week intervals for both series. After detrending, The Hungry Caterpillar's autocorrelations diminished appropriately, making both datasets suitable for ARIMA modeling with seasonal components (Figure 2).

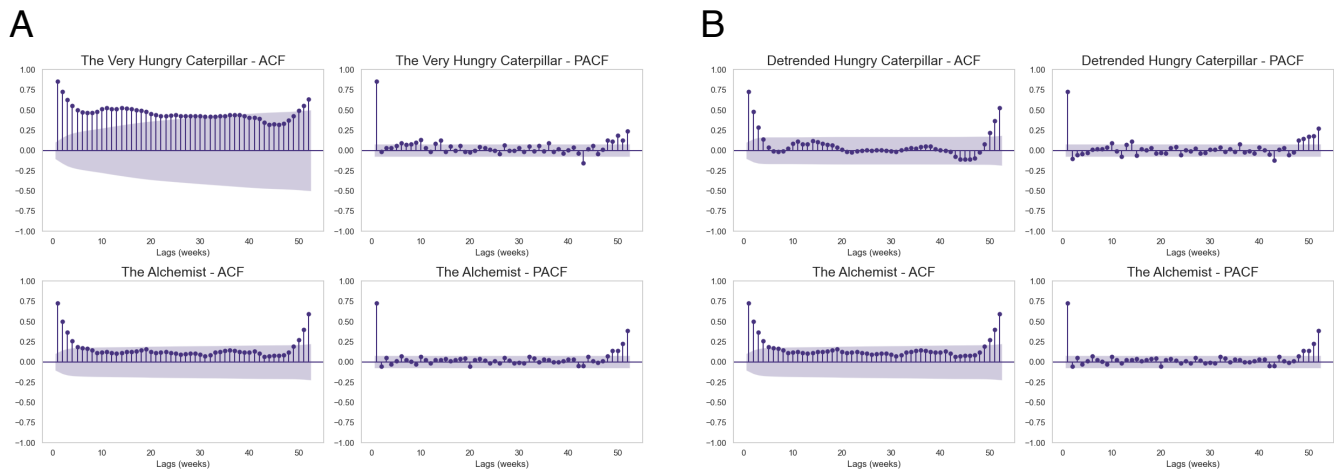


Figure 2: **Autocorrelation and Partial Autocorrelation Plots for Weekly Book Sales.** A) Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) for "The Very Hungry Caterpillar" and The Alchemist weekly sales data. B) Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) for "The Very Hungry Caterpillar" after detrending. Note that the Alchemist data was not detrended.

Model comparison revealed that Auto-ARIMA provided competitive performance against manually optimized specifications (Figure 3). For The Very Hungry Caterpillar, Auto-ARIMA (Model 4) outperformed the manually selected ARIMA (Model 5, AIC: 8041.90) in forecasting accuracy (MSE: 185,867 vs 192,415), with model selection prioritizing AIC over MSE based on superior visual forecast performance. Conversely, The Alchemist's manual Model 13 achieved superior forecasting performance (MSE: 24,021) compared to Auto-ARIMA (MSE: 40,017), despite higher AIC values. Residual diagnostics revealed assumption violations including non-normality and heteroscedasticity in both datasets, with The Alchemist showing additional autocorrelation issues. However, visual inspection demonstrated strong out-of-sample forecasting performance, highlighting the critical distinction between statistical diagnostic perfection and practical forecasting utility. This aligns with current practice emphasizing forecast accuracy over residual compliance, suggesting models should be evaluated primarily on predictive performance rather than strict adherence to diagnostic criteria.

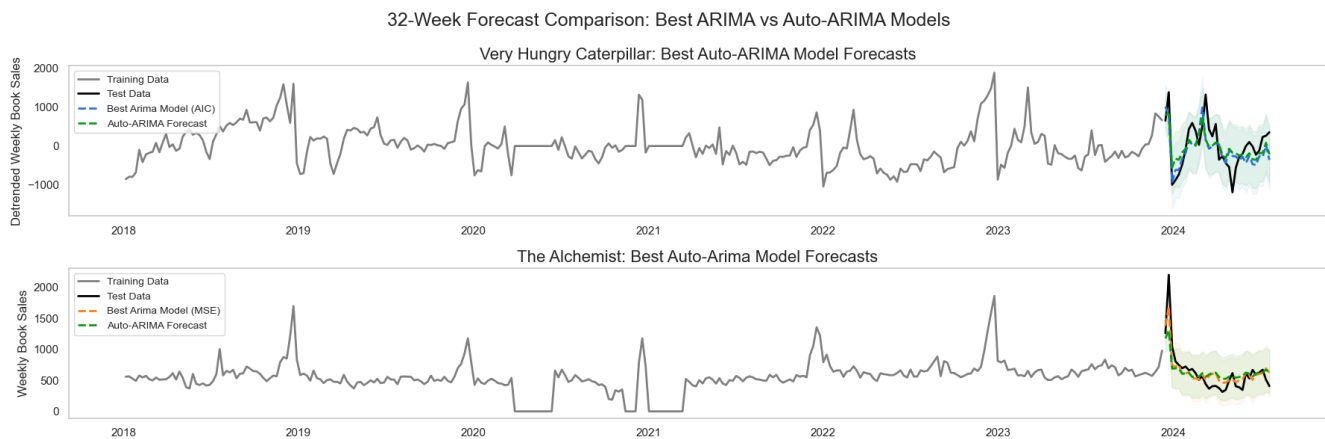


Figure 3: **Comparison of forecast accuracy between ARIMA and Auto-ARIMA models.** Top panels shows the 32-week out-of-sample forecast for "The Very Hungry Caterpillar" comparing the best ARIMA model selected by AIC with the Auto-ARIMA model. The bottom panel shows the 32-week out-of-sample forecast "The Alchemist," comparing the best Arima model based on MSE with the Auto-ARIMA model. Both panels highlight the model fits on the forecasts corresponding to the test period. Note only the last 4 years of data are displayed for clarity.

Given residual diagnostic violations in ARIMA models, LightGBM was tested with various preprocessing strategies including deseasonality, detrending, differencing, and their combinations (Figure 4). The recursive windowing implementation with TransformedTargetForecaster effectively handled non-stationary

characteristics across different preprocessing configurations. While LightGBM theoretically possesses superior capacity for capturing complex nonlinear patterns, visual inspection revealed that forecasting performance did not exceed ARIMA models for either dataset. This outcome suggests that the underlying sales patterns may be predominantly linear and seasonal, making simpler time series models more appropriate than complex machine learning approaches for this forecasting task.

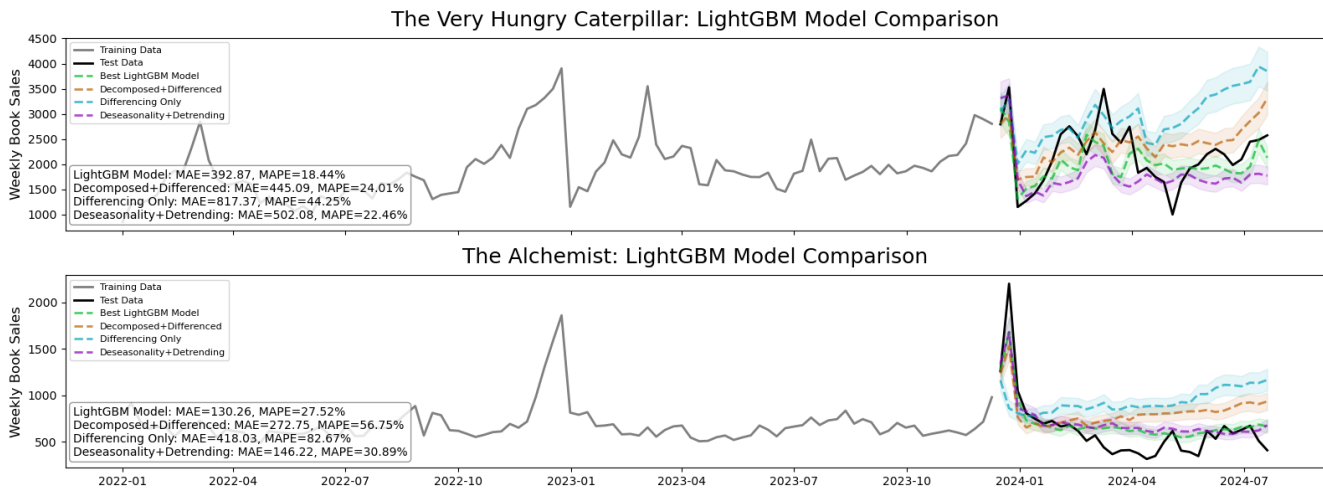


Figure 4: **LightGBM Model Comparison for "The Very Hungry Caterpillar" and "The Alchemist"**. A) LightGBM forecasting results for "The Very Hungry Caterpillar" weekly sales using different preprocessing strategies: Deseasonality + Detrending, Decomposed + Differenced, Differencing only, and Deseasonality + Detrending. B) LightGBM forecasting results for "The Alchemist" weekly sales using different preprocessing strategies: Deseasonality + Detrending, Decomposed + Differenced, Differencing only, and Deseasonality + Detrending. Forecasts were performed for the test period for comparisons. Performance in both data sets was measured by MAE and MAPE. Note only the last 2 years of data are displayed for clarity.

Building on the LightGBM analysis, LSTM models were subsequently evaluated with various preprocessing strategies (Figure 5). For The Very Hungry Caterpillar, deseasonalization achieved the best visual performance despite higher MAE (518.41) compared to detrending (436.93), though demonstrating superior MAPE (22.31% vs 26.07%). The Alchemist's LSTM models consistently underperformed across all preprocessing approaches, displaying predominantly linear patterns that failed to capture the actual sales dynamics. Despite occasionally achieving lower MAE/MAPE scores than LightGBM, LSTM models generally underperformed both traditional ARIMA and machine learning approaches. This suggests that the datasets' predominantly seasonal and linear characteristics are more effectively captured by simpler forecasting methodologies rather

than complex deep learning architectures.

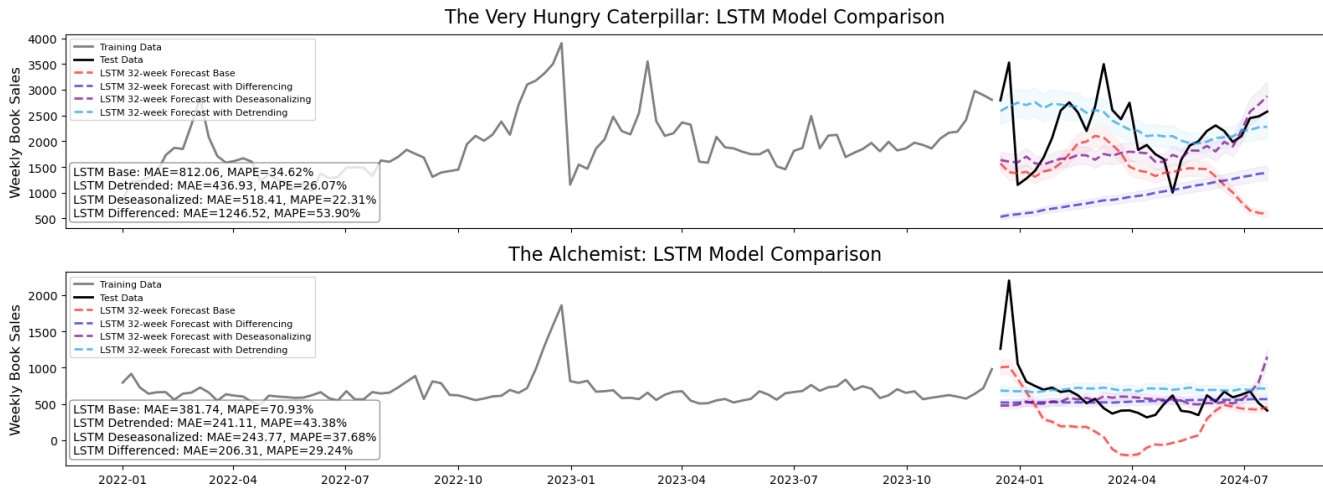


Figure 5: **LSTM Model Comparison for Weekly Sales Forecasts.** Top panel shows LSTM forecasting results for "The Very Hungry Caterpillar" weekly sales. Bottom panel presents LSTM forecasting results for "The Alchemist" weekly sales. In both datasets 32-week forecasts were generated using different LSTM preprocessing approaches: Base, Differencing, Deseasonalizing, and Detrending. Forecasts were performed for the test period for comparisons. Performance metrics (MAE and MAPE) were used to evaluate model accuracy. Note only the last 2 years of data are displayed for clarity.

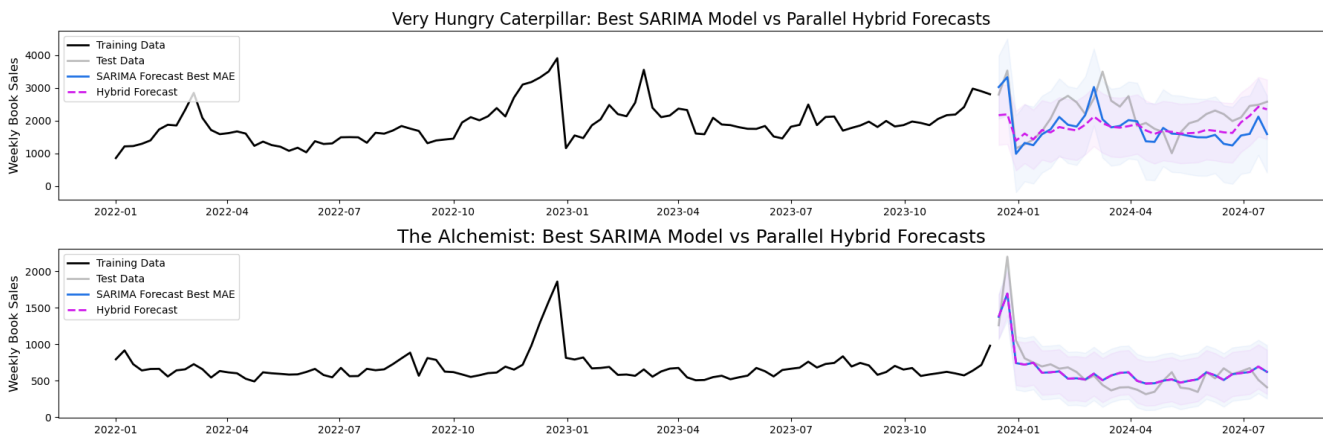


Figure 6: **Comparison of SARIMA vs Parallel Hybrid SARIMA-LSTM Forecasts for Weekly Sales.** Top panel shows the best SARIMA model forecast with the parallel hybrid SARIMA-LSTM forecast for "The Very Hungry Caterpillar" weekly sales from 2012 to 2024. Bottom panel presents the same comparison for "The Alchemist" weekly sales. Both panels show training and test data periods. Forecasts were performed for the test period for comparisons. Note only the last 2 years of data are displayed for clarity.

Considering the LSTM's limited success, hybrid SARIMA-LSTM models were explored to potentially combine ARIMA's statistical rigor with LSTM's nonlinear pattern recognition capabilities. Sequential hybrid approaches, which apply LSTM

to ARIMA residuals, proved inadequate due to insufficient residual structure in both datasets. Parallel hybrid configurations were then tested, running models independently before combining forecasts (Figure 6). While hybrid models demonstrated reduced residual variance, they followed observed patterns less closely than standalone SARIMA models. These findings indicate that classical time series methods are sufficient for these datasets, with deep learning adding minimal forecasting value while significantly increasing computational complexity and preparation time.

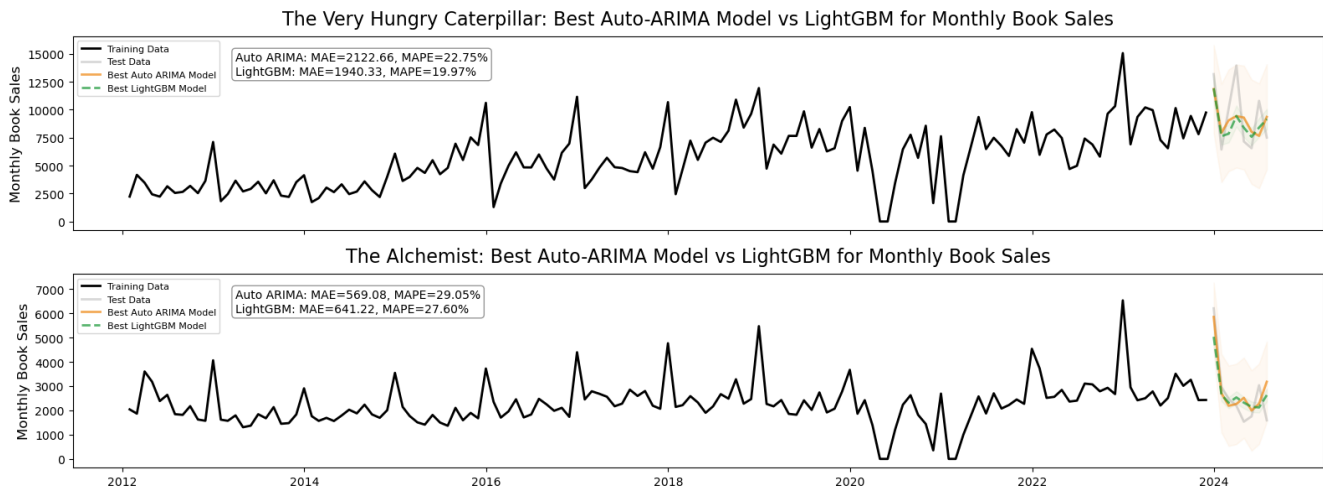


Figure 7: **Best Auto-ARIMA vs LightGBM Model Comparison for Monthly Book Sales.** Top panel compares the monthly forecast performance of the best Auto-ARIMA and LightGBM models for "The Very Hungry Caterpillar" showing training and test data with respective MAE and MAPE scores. Bottom panel presents the same comparison for "The Alchemist". Forecasts were performed for the test period for comparison.

To assess whether temporal resolution affects model performance, weekly data was resampled to monthly frequency for an 8-month forecasting evaluation (Figure 7). Both Auto-ARIMA and LightGBM demonstrated comparable performance at monthly aggregation, suggesting reduced temporal granularity diminishes complex modeling advantages. Monthly aggregation negatively impacted The Very Hungry Caterpillar forecasts, likely due to information loss from reduced granularity. Conversely, The Alchemist showed improved monthly performance compared to the very hungry caterpillar set, reflecting differences in underlying pattern complexity between datasets. These findings indicate that optimal temporal resolution depends on dataset characteristics, with monthly aggregation not universally improving prediction accuracy compared to weekly forecasts.

5 Conclusions

This time series analysis demonstrated that classical Auto-ARIMA models consistently outperformed machine learning and deep learning approaches for forecasting “The Very Hungry Caterpillar” and “The Alchemist” sales. While both datasets exhibited strong seasonality, preprocessing responses varied significantly. The Very Hungry Caterpillar benefited from detrending due to its trend-stationary nature, whereas The Alchemist remained robust across transformations.

LightGBM models showed competitive performance but failed to exceed ARIMA accuracy, while LSTM models consistently underperformed despite their theoretical nonlinear capabilities. Hybrid SARIMA-LSTM models reduced residual variance but produced forecasts more distant from actual patterns than standalone classical methods. Monthly aggregation did not improve prediction accuracy compared to weekly forecasts. The findings emphasize that model complexity does not guarantee superior forecasting performance for datasets with predominantly linear and seasonal characteristics.

Actionable Insights for these two books:

- Implement title-specific forecasting strategies based on sales characteristics—apply detrending for books showing growth trends (like “The Very Hungry Caterpillar”) while maintaining standard seasonal adjustments for stable sellers (like “The Alchemist”).
- Optimize inventory management using 32-week Auto-ARIMA forecasts to reduce stockouts and overstock situations.
- Focus weekly sales forecasting over monthly aggregation to capture demand volatility.

6 References

1. Notebook 1: https://github.com/rgrandym/data_science_cambridge/tree/main/course/reports/course_3#:~:text=Grandy_Rodrigo_CAM_C301_Week_9_10_Topic_project_nootebook1.ipynb
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